

Sound APIs



Games today demand an interactive experience by definition. When a character speaks in a game, its voice should pan around the listener depending on the position of the speaker, or a gunshot should sound different in a closed room as compared to an open space. These demands have led soundcard manufacturers and game developers to produce and use what are known as sound APIs. A game would call on the soundcard and with the help of the necessary API it would deliver the requisite effect. If a soundcard supported DirectSound, it was labeled accordingly and the same logic extends to the other APIs.

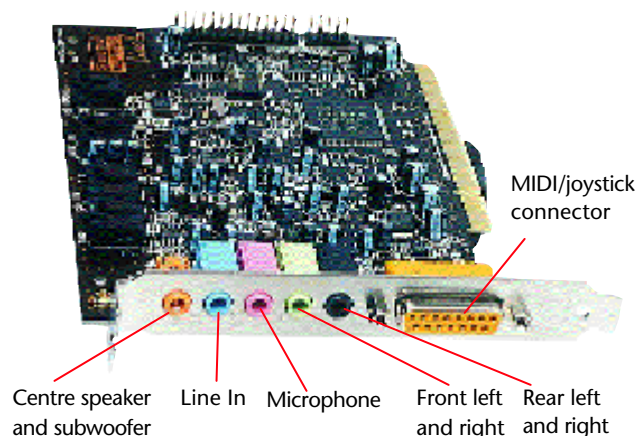
DirectSound3D (DS3D) was introduced in DirectX 3.0 and allowed developers to place a sound anywhere in 3D space. Although DS3D allowed for 3D sound effects, it did not offer much in the way of simulating a particular environment. Thus a gunfight in a game would sound much the same in an auditorium and a cave. This mantle was taken up by other APIs such as Creative Labs' **Environmental Audio Extension (EAX)** and **Aureal's A3D**. EAX took the relatively easy route and with the help of reverberations and echoes simulated the properties of a place, rather than the place itself. Thus under EAX, the echo in a cave would be different from the sound reverberations in, say, a corridor. A3D went all out and decided to completely simulate a locale in question. This was akin to creating a virtual world in 3D using a program like 3D Studio Max, but the treatment here was aural rather than visual.

There are other APIs as well, noteworthy amongst them being **I3DL2**. It is a collaborative effort that is aimed at extending the capabilities of Creative Labs' EAX open reverberation standard. It defines the space where the listener is located, takes advantage of the reverberation cues in the current environment and renders the muffling effects of obstacles inside environments or partitions between environments.

Creative Audigy Platinum outshone the others in this department, providing drivers for all major operating systems.

For those using their PCs as machines to create and mix music, the number of MIDI channels supported and SoundFont support are critical features. MIDI channels are nothing but the number of instruments or vocals that a soundcard can support through a synthesiser. Out of the lot, only Creative's Vibra 128 supported a meagre 16-channels, the rest offered an excellent 48-channel solution. Although home users would hardly find any difference while listening to MP3s or playing games, for a professional sound artist this could be the most crucial factor. SoundFonts allows users to create their own sample music and play them back, create some beats, or even grab some music from a digital audio file. It also lets you play most of the known musical instruments available, through your PC. Only the Audigy Platinum, Extigy and the Sound Blaster Live! DE 5.1 sported this feature. If you are looking for a card that will let you record your own audio, keep the following in mind: the more the recording-depth and bit rate that a soundcard offers, the better the quality of sound that can be recorded. Creative's Extigy was the clear leader here. It offered a high 24-bit depth at a 96 KHz sampling

Soundcard Connectors Demystified



frequency for both playback and recording, proving to be the clear choice for the professionals. The Audigy had a playback depth of 24 bits but the recording was at the normal 16-bit depth. The other cards lagged behind with 16-bit recording depth and 48 KHz sampling rate.

Finally, we come to the often-overlooked feature of connectors. Most cards today sport all the necessary connectors like Line Out, Microphone Out, MIDI and joystick ports, etc. If you are looking for a normal two-speaker solution, you should not have any problems with any of the available cards. However, for those looking at building a home theatre system (with 5.1 speakers), the connectors become all the more important. An S/PDIF or any other Digital Out connection is a must here as it provides a pure digital signal without any interference. It can also act as a pass through in case the soundcard doesn't have any Dolby Digital decoding capability—the speaker's Dolby decoder unit will then decode the digital signal. These connectors were only available in the premium soundcards available from Creative: the Audigy Platinum and the Extigy. The others like Zoltrix's Nightingale Pro 6 did provide six-channel output, but did not feature an S/PDIF Out connector.

A unique feature to note is that Creative Audigy Platinum has a daughter Audigy drive. The external drive gives better access to controls and comes with an SB1394 (FireWire) connector for connecting ultra fast playback and recording devices. Our opinion is that instead of FireWire, USB 2.0 support would have been a better proposition, as there are more devices supporting this standard.



The Audigy drive packs a lot of connectors on the front panel

The Extigy is aimed at notebook users and was the only external 'soundcard' available. The Sound Blaster Extigy provides hassle-free installation by easily plugging into any available USB port on your PC or notebook. But to our dismay, the Extigy only had support for USB 1.1, which can become a bottleneck as shown in our CPU utilisation tests below.

Performance

To find out the best amongst the available soundcards, we used a battery of real world as well as benchmark applications. A